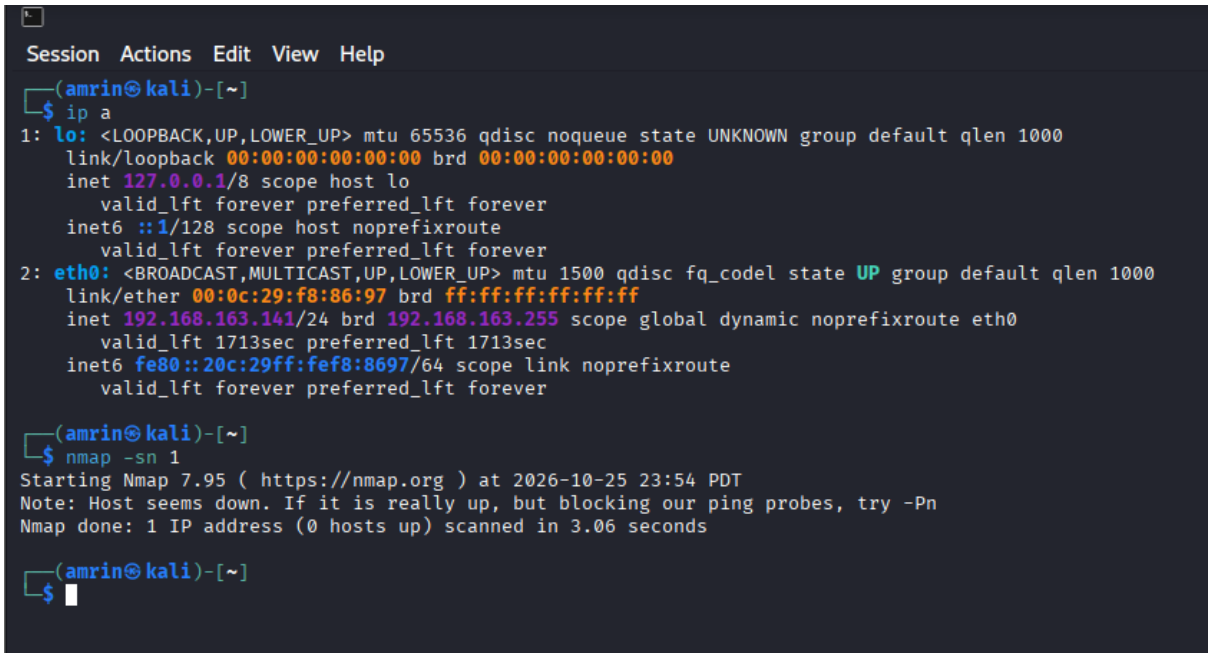
1. Install Nmap on your system and use it to scan your local Wi-Fi network for all connected devices. List the IP addresses and device names (if available) found in the scan.

Ans-

Step 1: Install Nmap

Step 2: Find your Wi-Fi IP address and subnet

Step 3: Scan : nmap -sn 192.168.163.141/24



2. Use the 'arp -a' command in your terminal or command prompt to display the current ARP table. Identify at least two devices on your network and write down their IP and MAC addresses.

Ans-

```
C:\Users\amrin>arp -a

Interface: 192.168.163.1 --- 0x5
  Internet Address      Physical Address      Type
  192.168.163.254       00-50-56-fe-41-31     dynamic
  192.168.163.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.110.1 --- 0xf
  Internet Address      Physical Address      Type
  192.168.110.254       00-50-56-e6-07-c9     dynamic
  192.168.110.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 172.16.0.122 --- 0x12
  Internet Address      Physical Address      Type
  172.16.0.1            3c-78-95-cc-9e-92     dynamic
  172.16.0.104          10-f6-0a-c9-57-6b     dynamic
  172.16.0.109          e6-b1-06-25-48-51     dynamic
  172.16.0.115          56-56-77-19-01-86     dynamic
  172.16.0.255          ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\amrin>
```

3.Perform a port scan on your own computer using Nmap to find which common ports (like 22, 80, 443) are open. Summarize your findings in a table with port number, service, and status (open/closed).

Ans-

```
┌──(amrin㉿kali)-[~]
└─$ nmap -p 22,80,443 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-10-25 23:58 PDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00015s latency).

PORT    STATE  SERVICE
22/tcp  closed ssh
80/tcp  open   http
443/tcp closed https

Nmap done: 1 IP address (1 host up) scanned in 0.07 seconds

┌──(amrin㉿kali)-[~]
└─$
```

**4.Simulate a Zomato-style scenario: Imagine you are a security tester for a food delivery app. Use Nmap to check if the test server at 'scanme.nmap.org' has any open ports that could be vulnerable. List any ports found and suggest what kind of services might be running on them.

Hint: Use the command 'nmap scanme.nmap.org' and research the common services for the ports you find.**

Ans-

Step 1: Run the basic scan

nmap scanme.nmap.org

Wait until Nmap finishes and displays the port table.

Step 2: Read the port results

You may see output in this format:

PORT STATE SERVICE

22/tcp open ssh

80/tcp open http

The above is illustrative; it is not a claim about the server's current ports.

Step 3: Understand common ports

Port 22 — SSH

Secure remote login and administration.

Port 80 — HTTP

Web traffic that is not encrypted by HTTP itself.

Port 443 — HTTPS

Web traffic typically protected using TLS encryption.

Port 3306 — MySQL

Commonly used for MySQL database connections.

A port number alone does not prove which application is running or whether it is vulnerable.

Step 4: Record your findings

Port found	State	Possible service	Security recommendation
Actual port	Actual state	Based on Nmap output	Review exposure and access controls
Actual port	Actual state	Based on Nmap output	Verify updates and secure configuration

**5.Refactor your Nmap scan command to only show open ports and their services, hiding all closed ports from the output.

Constraint: Use the '-open' or '-sV' flag in your command to filter results.**

Ans-

